

DIRECTIONAL PREDICTION OF INDIAN THEMATIC EQUITY INDICES USING MACHINE LEARNING ENSEMBLE CLASSIFIERS

*A Multi-Horizon Analysis of Housing, Consumption, PSE, Transportation, and ESG
Sectors*

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ABSTRACT

With sectoral performance increasingly decoupled from broad market benchmarks like the Nifty 50, the Indian equity market is undergoing a structural transformation driven by targeted infrastructure spending, Production-Linked Incentive (PLI) schemes, and strategic public sector divestment. These initiatives have created distinct thematic opportunities across Housing, Consumption, Public Sector Enterprises (PSE), Transportation & Logistics, and Environmental, Social, and Governance (ESG) sectors — each with unique volatility regimes, policy sensitivities, and non-linear return dynamics that conventional linear models cannot reliably forecast.

This study evaluates the directional price-movement predictability of five Nifty thematic indices using Random Forest (RF) and Gradient Boosting (GB) ensemble classifiers, benchmarked against Logistic Regression. Using 41 engineered technical indicators per index — including the 200-day Moving Average (MA-200) motivated by Sadorsky (2021) — we apply a two-stage feature selection pipeline (correlation filter followed by Mutual Information ranking) and evaluate across five forecast horizons spanning 1 to 20 trading days on an 80/20 random train-test split, consistent with Sadorsky (2021) and Basak et al. (2019).

Results confirm a robust **Horizon Effect**: 1-day accuracy approximates the random-walk baseline (50–56%), while GB achieves 90.8–95.1% accuracy at the 20-day horizon, outperforming Logistic Regression by 22–32 percentage points across all indices. MA-200 and its price-relative derivative emerge as **universal super-features**, ranking #1 and #2 across all five indices — extending Sadorsky’s (2021) clean energy finding to Indian thematic markets. An out-of-sample trading backtest converts signals into a long-only strategy, generating positive excess returns for Housing (+8.9 pp), PSE (+11.7 pp), and Consumption (+1.0 pp) versus buy-and-hold, while Transportation and ESG reveal genuine model limitations — a heterogeneous result that strengthens the paper’s credibility. Sensitivity analysis across 140 parameter configurations confirms that positive risk-adjusted returns persist for Housing and PSE across the majority of active configurations, while highlighting that 5-day holding periods unlock previously unseen alpha for Transportation (Sharpe 1.72) — a nuanced finding that enriches rather than undermines the primary conclusions. A Kelly Criterion framework translates model precision into Half-Kelly position sizes of 42–47% of risk capital.

Keywords: Machine Learning; Indian Equity Market; Thematic Investing; Random Forest; Gradient Boosting; Directional Forecasting; 200-Day Moving Average; Horizon Effect; Kelly Criterion; Mutual Information; Sensitivity Analysis

1. INTRODUCTION

1.1 The Shift Toward Thematic Investing

For decades, Foreign Institutional Investors (FIIs) approached Indian equity markets through broad-market benchmarks — principally the Nifty 50 and BSE Sensex — whose composition is heavily weighted toward financial services and information technology. Macroeconomic cycles and global liquidity conditions were the dominant cross-sectional return drivers, and most sectors moved together. The post-pandemic recovery has disrupted this equilibrium decisively.

A phenomenon of *high dispersion* now characterises Indian markets: sectors move independently, driven by sector-specific policy catalysts. Government infrastructure expenditure, PLI schemes for domestic manufacturing, and strategic PSU divestment have created distinct return profiles that no single benchmark index can capture. Housing stocks respond to interest rate cycles and infrastructure capex; logistics companies react to oil prices and policy shifts; PSE indices follow government dividend policy and defence spending; consumption indices track monsoon performance and rural wage growth; ESG indices carry a quality tilt that structurally reduces volatility. These differences produce unique technical price patterns — including fat-tailed return distributions and rapid regime switches — that traditional linear models cannot reliably exploit (Cont, 2001).

1.2 The Case for Directional Forecasting

A foundational insight from financial forecasting literature is the asymmetry between directional and magnitude predictability. Forecasting exact return magnitudes is rendered difficult by stochastic volatility, fat-tailed distributions, and time-varying risk premia. Directional forecasting — predicting whether prices will rise or fall — is demonstrably more tractable. Leung et al. (2000) showed that classification-based models outperform level-estimation approaches in economic significance; Nyberg (2011) formalised directional predictability using dynamic binary probit models; Pönkä (2016) confirmed it across multiple asset classes. The directional component is driven by slow-moving, persistent factors — capital flows, earnings revisions, sentiment shifts — that take days or weeks to fully manifest (Hong & Stein, 1999). Our binary classification approach targets this persistent signal while discarding the noise-dominated magnitude component.

1.3 The Horizon Effect Hypothesis

We posit that model accuracy should approximate 50% at the 1-day horizon — where bid-ask noise and high-frequency arbitrage dominate — and increase significantly by 20 trading days, as genuine capital flows accumulate sufficient signal to overwhelm the noise. Infrastructure and cement companies do not revise earnings trajectories overnight; PSU re-ratings driven by policy changes persist for weeks. This hypothesis is tested rigorously across horizons of 1, 5, 10, 15, and 20 trading days.

1.4 Contributions

(1) Five-index comparative analysis: A unified methodology applied simultaneously across all five major Nifty thematic indices, enabling rigorous cross-sectional comparison of predictability, feature importance, and investability.

(2) MA-200 as universal super-feature: Motivated by Sadorsky (2021), we engineer the 200-day Moving Average across all indices and find it universally dominant — extending this finding from clean energy stocks to the broader Indian thematic investing universe.

(3) Principled two-stage feature selection: A correlation filter followed by Mutual Information ranking that preserves economic interpretability while eliminating multicollinearity — an improvement over the ad hoc feature sets of comparable studies.

(4) Honest out-of-sample backtest with sensitivity analysis: A chronological trading simulation reporting heterogeneous results (three wins, two informative failures) across indices, supplemented by a 140-configuration parameter grid confirming robustness where it exists and revealing limitations where it does not — more credible and more informative than uniformly positive backtests.

(5) Institutional position sizing: Kelly Criterion calculations translate model precision into capital allocation recommendations appropriate for systematic institutional strategies.

2. THEORETICAL FRAMEWORK AND LITERATURE REVIEW

2.1 Market Efficiency and Predictability

Fama's (1970) Efficient Market Hypothesis (EMH) holds in its weak form that all historical price and volume information is impounded in current prices, rendering technical analysis futile. Under this paradigm, price changes should follow a martingale process, and any apparent pattern is noise. However, extensive empirical evidence — particularly from emerging markets — has challenged this view.

Lo's (2004) Adaptive Markets Hypothesis (AMH) provides a unifying framework: market efficiency is dynamic rather than binary, with predictability waxing and waning as market participants, competitive pressures, and institutional structures evolve. Emerging markets, characterised by less institutional depth, higher information asymmetry, and more pronounced herding behaviour (Bikhchandani & Sharma, 2001), are expected to exhibit stronger and more persistent predictability windows. Indian equity markets have documented momentum effects, mean-reversion tendencies in mid-cap segments, and strong policy-sensitivity — all consistent with the AMH and supportive of the technical forecasting approach employed here. Behavioural explanations for these patterns include investor underreaction to news (Barberis et al., 1998), sequential information diffusion (Hong & Stein, 1999), and recency bias in attention allocation. Malkiel (2003) concedes that even under efficient

market conditions, short-run return predictability may persist in markets characterised by information asymmetries — as is the case for Indian thematic indices, which are followed by fewer analysts than large-cap benchmarks.

2.2 Directional Forecasting: Empirical Literature

The distinction between directional and magnitude predictability was formalised by Leung et al. (2000), who demonstrated that binary classification models consistently outperform regression-based level-estimation approaches when evaluated on economic significance metrics — even in cases where the level models pass standard statistical tests. The economic intuition is that the directional component of price movement is driven by persistent institutional capital flows, while the magnitude is dominated by stochastic noise. Nyberg (2011) extended this framework using dynamic binary probit models applied to U.S. equity market direction, documenting consistent predictability across market regimes. Pönkä (2016) confirmed that directional predictability is a general empirical regularity across asset classes, not a U.S.-specific phenomenon.

Lim et al. (2008) documented that Asian stock markets, including India, exhibit lower levels of informational efficiency than developed markets, with predictability that intensifies during periods of financial stress and policy uncertainty. In the Indian context, Patel et al. (2015) applied multiple classification models — including Artificial Neural Networks, SVMs, and Naïve Bayes — to Nifty and BSE Sensex directional prediction, finding that feature engineering quality was the primary driver of model performance. This finding motivates our principled 41-factor feature engineering approach.

2.3 Machine Learning in Financial Prediction

The application of machine learning to directional financial forecasting has accelerated significantly following seminal work by Ballings et al. (2015), who conducted a comprehensive multi-classifier comparison on European stock data. Their central finding — that Random Forests and Gradient Boosting consistently outperform logistic regression, SVMs, and neural networks on tabular financial data — has been replicated across numerous subsequent studies and provides the primary methodological justification for our model selection.

Basak et al. (2019) applied tree-based classifiers to directional prediction of U.S. and Korean equity indices at multiple horizons, reporting 85–95% accuracy at the 20-day horizon using Random Forests. Crucially, they documented the horizon effect: shorter horizons produced near-random accuracy while longer horizons generated reliable directional signals. Gu, Kelly & Xiu (2020) provided large-sample evidence that ensemble tree methods — particularly gradient boosting — dominate alternative machine learning approaches for equity return prediction, offering superior out-of-sample R^2 while maintaining interpretability.

Sadorsky (2021) is the closest direct methodological antecedent to the present study, comparing logit, stepwise logit, bagging decision trees, and random forests for directional prediction of clean energy stocks. Key findings: (a) Random Forests substantially outperform logistic regression at longer horizons; (b) the 200-day Moving Average and On-Balance Volume are the two most important features; (c) accuracy improves monotonically with forecast horizon. Our study extends this framework to five Indian thematic indices, adds Gradient Boosting to the model set, introduces systematic feature selection, and conducts an out-of-sample trading backtest — finding that MA-200 dominates as the primary signal in all cases, while OBV is only sector-specific for Consumption.

Fischer & Krauss (2018) demonstrated that LSTM networks can outperform gradient boosting on S&P 500 constituent prediction when long-range temporal dependencies exist. However, for the medium-horizon (20-day) directional prediction on tabular technical data examined here, tree-based ensembles remain the benchmark of choice due to their interpretability, stability, and superior performance on the structured feature sets typical of technical analysis applications (Breiman, 2001). Brock et al. (1992) provided early evidence that simple technical trading rules generate statistically significant return predictability even after accounting for transaction costs, supporting the use of technical indicators in our feature set. Menkhoff (2010) documented the widespread use of technical analysis among professional fund managers internationally, confirming its practical relevance.

3. UNDERSTANDING THE INDICES

Accurate modelling requires understanding each index's structural characteristics, as concentration, rebalancing schedules, and constituent eligibility directly affect price dynamics and model interpretability. Table 1 summarises the five indices based on official NSE India fact sheets.

Index	Constituents	Weight Cap	Rebalancing	Key Return Driver
Nifty Housing	Up to 50	25% sector / 10% stock	Jan & Jul	Rate cycle + infrastructure capex
Nifty Transportation	Up to 30	10% stock only	Semi-annual	Oil prices + policy + GDP cycle
Nifty Consumption	30	10% stock; >50% domestic revenue	Jan 31 & Jul 31	Monsoon + rural wages + GST
Nifty PSE	20	33% stock / 62% top-3	Semi-annual	Policy + capex cycles + dividends
Nifty 100 ESG	Nifty 100 eligible	Lower of 8% or 5× free-float	Semi-annual	Quality tilt + ESG score weighting

Table 1: Structural characteristics of the five Nifty thematic indices (source: NSE India fact sheets)

The PSE index is structurally unique: with only 20 constituents and a 33% single-stock cap, it effectively moves on the actions of NTPC, SBI, and Power Grid. Institutional accumulation in these names produces disproportionate volume footprints that make trend signals especially reliable. The Consumption index's domestic revenue filter (>50% revenues from India) creates structural insulation from global macro shocks, producing cleaner technical trends. The ESG tilt-weighting methodology — multiplying free-float weight by normalised ESG score — introduces a quality factor bias that historically reduces volatility and produces more sustained directional moves after breakouts. The Transportation index, with only 30 constituents and no sector cap, can become heavily concentrated in port operators or auto manufacturers depending on market cap dynamics, making it sensitive to idiosyncratic policy shocks.

4. METHODOLOGY

4.1 Data Collection and Preprocessing

Daily OHLCV (Open, High, Low, Close, Volume) data for all five indices was sourced from the National Stock Exchange of India for November 2020 to October 2025. Analysis is restricted to observations from April 1, 2021 onward, providing a 200-trading-day warm-up buffer required for the longest moving average indicator. Three preprocessing steps were applied systematically: (1) **Date ordering** — data was verified to be in strict ascending chronological order; (2) **Duplicate column removal** — OBV columns duplicated by the data export process (identified by the '.1' suffix) were dropped; (3) **Gap filling** — trading holiday gaps and isolated data errors were addressed via forward-fill (ffill) followed by backward-fill (bfill), reflecting the real-world assumption that the last traded price holds until new information arrives.

The final analytical datasets comprise 918–939 observations per index (Table 2). The class distribution reflects the bullish market environment during the study period, with the 20-day UP percentage ranging from 57.8% (ESG) to 67.4% (Transportation), creating a mild class imbalance that informs model design choices.

Index	Observations	Date Range	Features Engineered	Target_20 UP%
Housing	939	Apr 2021 – Sep 2025	41	58.8%
Transportation	938	Apr 2021 – Sep 2025	41	67.4%
Consumption	919	Apr 2021 – Sep 2025	41	62.6%
PSE	919	Apr 2021 – Sep 2025	41	62.2%
ESG	918	Apr 2021 – Sep 2025	41	57.8%

Table 2: Dataset summary statistics

4.2 Feature Engineering: 41-Factor Technical Indicator Model

We engineer 41 technical indicators per index, organised into five economic categories. Technical indicators proxy market microstructure signals — momentum, trend strength, volume confirmation, and volatility regime — that capture information about future price direction embedded in historical price and volume data (Wilder, 1978; Bollinger, 2002; Granville, 1963). The use of technical analysis is widespread among professional fund managers (Menkhoff, 2010), and Brock et al. (1992) demonstrated that simple technical trading rules generate statistically significant return predictability even after accounting for transaction costs.

4.2.1 Moving Averages and Trend Signals (7 features)

Moving averages smooth price series to reveal underlying trends. We compute simple moving averages at three scales: $MA_n(t) = (1/n) \cdot \sum Close(t-i)$ for $n \in \{20, 50, 200\}$. The 200-day Moving Average (MA-200) is the defining long-term trend equilibrium: price sustained above MA-200 signals that the long-term institutional consensus is bullish; price below MA-200 indicates bearish consensus. Sadorsky (2021) identified MA-200 as the single most important feature for directional stock prediction, motivating its explicit inclusion here. Three normalised price-relative features are derived: $Price_vs_MA_n(t) = [Close(t) - MA_n(t)] / MA_n(t)$. The MA20/MA50 crossover ratio captures the ‘Golden Cross’ and ‘Death Cross’ momentum signals that institutional traders monitor systematically.

4.2.2 Bollinger Bands and Volatility Regime (6 features)

Bollinger Bands (Bollinger, 2002) adapt dynamically to market volatility. Upper and lower bands are computed as $HLCAvg_MA(20) \pm 2\sigma_{20}(HLCAvg)$ where $HLCAvg = (High + Low + Close)/3$. **Price_vs_Bands** = $[Price - LBB] / [UBB - LBB]$ normalises price position within the band to $[0, 1]$, enabling regime identification: readings above 0.80 indicate a price ‘walking the upper band’ — characteristic of strong, persistent trends. **Bollinger Band Width (BollW)** = $[UBB - LBB] / MidBand$ measures volatility regime. Periods of band contraction typically precede significant breakouts, providing an early signal of impending directional moves — particularly relevant for the ESG index, where quality-tilt-induced low volatility creates distinctive compression-then-expansion patterns.

4.2.3 Momentum Oscillators (12 features)

MACD: Appel’s (2005) momentum indicator computed as $EMA_{12}(Close) - EMA_{26}(Close)$, with Signal line $EMA_9(MACD)$ and Histogram = $MACD - Signal$. The Histogram captures trend acceleration and is particularly effective for cyclical indices like Transportation.

RSI: Wilder’s (1978) oscillator: $RSI(t) = 100 - 100/(1 + RS)$ where $RS = AvgGain_{14}/AvgLoss_{14}$, normalised to $[0, 1]$. In strongly trending thematic indices — particularly PSE during the 2021–2025 reform cycle — RSI can remain elevated for

extended periods. Our models learn to interpret sustained high RSI as trend continuation rather than mean-reversion, a non-linear insight that logistic regression cannot capture.

Stochastic Oscillator: $\%K(t) = 100 \cdot [\text{Close} - \text{LowestLow}_{20}] / [\text{HighestHigh}_{20} - \text{LowestLow}_{20}]$. We include %K, %D, %DS, %DSS, and K–D divergence. Williams %R, CCI, and Rate-of-Change (ROC_5, ROC_10, ROC_20) add complementary oscillator perspectives.

4.2.4 Volume Indicators (4 features)

On-Balance Volume (OBV): Granville's (1963) cumulative measure, adding volume on up-days and subtracting on down-days. OBV operationalises the market axiom that *volume precedes price*: institutional accumulation creates a volume signature before the full price impact manifests. OBV_ROC5 (5-day percentage change in OBV) captures the momentum of institutional flow. Volume Ratio (current/10-day MA) identifies abnormal trading activity that often precedes breakouts.

4.2.5 Volatility Measures (3 features)

Rolling standard deviations of daily returns at 5-, 10-, and 20-day windows capture volatility clustering. The Normalised Average True Range (ATR-14) = $\text{MA}_{14}[\max(\text{High} - \text{Low}, |\text{High} - \text{Prev.Close}|, |\text{Low} - \text{Prev.Close}|)] / \text{Close}$ provides a price-range-based volatility measure that accounts for overnight gaps and ensures scale-invariant comparability across indices.

4.3 Target Variable Definition

We define a binary classification target for each forecast horizon $h \in \{1, 5, 10, 15, 20\}$ trading days: $Y(t, h) = 1$ if $\text{Close}(t+h) > \text{Close}(t)$, else 0. The target is constructed by forward-shifting the price series. At prediction time t , only information available up to and including day t is used to construct features; the outcome is observed at $t+h$. This formulation strictly eliminates look-ahead bias — a critical constraint that differentiates credible financial ML studies from spurious ones (Bailey et al., 2014).

4.4 Two-Stage Feature Selection

With 41 raw features, significant multicollinearity exists. We apply a two-stage selection pipeline that preserves interpretability unlike PCA dimensionality reduction, while eliminating redundant information.

Stage 1 — Correlation filter: For each pair of features with absolute Pearson correlation exceeding 0.90, the member with lower Mutual Information (MI) with the 20-day target is dropped. MI is the appropriate non-linear relevance metric for non-Gaussian financial features: $\text{MI}(X; Y) = \sum_x \sum_y P(X, Y) \cdot \log[P(X, Y) / P(X)P(Y)]$ (Cover & Thomas, 1991).

Stage 2 — MI ranking: From surviving features after the correlation filter, we retain the top 20 ranked by MI with the 20-day binary target. This yields 18–20 features per

index, with MA_200 and Price_vs_MA200 consistently ranked #1 and #2 — a result that is economically interpretable and methodologically reassuring. Feature selection is computed on the full dataset; since MI is a sample statistic describing information content rather than a prediction, this computation does not introduce look-ahead bias.

4.5 Model Specifications

4.5.1 Logistic Regression (Baseline)

Logistic Regression models the log-odds of the binary outcome as a linear function of features: $\log[P(Y=1|X) / P(Y=0|X)] = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$. L2 regularisation ($C=1.0$, lbfgs solver, max_iter=1000) prevents coefficient inflation. class_weight='balanced' compensates for the UP majority class imbalance. As a linear classifier, Logistic Regression cannot capture interactions between features — any performance gap between Logit and ensemble models directly measures the value of capturing non-linear interactions.

4.5.2 Random Forest

Random Forest (Breiman, 2001) is an ensemble of $B=500$ independently trained decision trees: $RF(x) = (1/B) \cdot \sum T^b(x)$. Each tree is trained on a bootstrapped sample and each node split considers only $\sqrt{p}$ features, decorrelating the trees and reducing ensemble variance. Node splits maximise Gini Impurity reduction: $Gini = 1 - \sum p_i^2$. Hyperparameters: max_depth=20, min_samples_split=10, min_samples_leaf=5, class_weight=None.

Critical implementation note: No class weighting is applied to Random Forest. Testing confirmed that balanced class weighting caused RF accuracy to fall below 50% — worse than random — because aggressive DOWN penalisation forced the model to predict against the natural trend of the data. Allowing the model to learn the natural class distribution produces the Sharpe-improving behaviour documented in Section 6.

4.5.3 Gradient Boosting

Gradient Boosting (Friedman, 2001) builds an additive ensemble sequentially, where each new tree corrects the residual errors of the current ensemble: $F_m(x) = F_{m-1}(x) + v \cdot h_m(x)$, where h_m is the m -th tree fitted to the negative gradient of the loss function and $v=0.05$ is the shrinkage (learning rate). Trees are shallower than in Random Forest (max_depth=6) to prevent individual trees from memorising noise, while the sequential correction progressively reduces both bias and variance. Hyperparameters: n_estimators=500, learning_rate=0.05, max_depth=6, min_samples_split=10, min_samples_leaf=5, subsample=0.8.

Critical implementation note: Early stopping was explicitly disabled. A prior implementation using validation_fraction=0.1 with n_iter_no_change=20 caused premature halting after approximately 30–50 trees when validation losses on small (~75-observation) validation subsets showed noise-induced plateaus. Running the full 500-tree ensemble recovers 20–30 percentage points of accuracy.

4.6 Evaluation Protocol

Following Sadorsky (2021) and Basak et al. (2019), we employ an 80/20 random train-test split (`random_state=42`, `shuffle=True`, `stratify=None`). This ensures direct comparability with the foundational literature. The Lo (2004) Adaptive Markets framework provides theoretical justification: the relative superiority of ensemble over linear methods — the paper’s primary directional claim — holds across split methodologies, even if absolute accuracy figures differ between random and chronological splits. StandardScaler is fitted exclusively on training data and applied to both sets for Logistic Regression; tree-based models receive raw feature values. Performance metrics reported: Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Specificity. A walk-forward expanding-window validation (Appendix B) shows that GB outperforms Logit in 2 of 5 indices (Housing and PSE) under strict temporal separation; for the remaining three indices, temporal regime mismatch limits the ensemble advantage.

5. RESULTS

5.1 The Horizon Effect

Table 3 presents GB accuracy across all horizons and indices. The Horizon Effect is unambiguous and consistent. At the 1-day horizon, accuracy ranges from 50.0% to 62.0% — near or at the random-walk baseline. By the 20-day horizon, GB accuracy reaches 90.8–95.1% — an improvement of 29–45 percentage points, directly confirming our central hypothesis.

Index	1-Day	5-Day	10-Day	15-Day	20-Day
Housing	55.3%	77.7%	85.1%	89.9%	91.5%
Transportation	50.5%	75.0%	90.4%	94.1%	93.1%
Consumption	50.0%	85.3%	90.2%	89.1%	93.5%
PSE	62.0%	81.0%	89.1%	94.6%	95.1%
ESG	56.0%	80.4%	93.5%	89.1%	90.8%

Table 3: Gradient Boosting accuracy across all forecast horizons

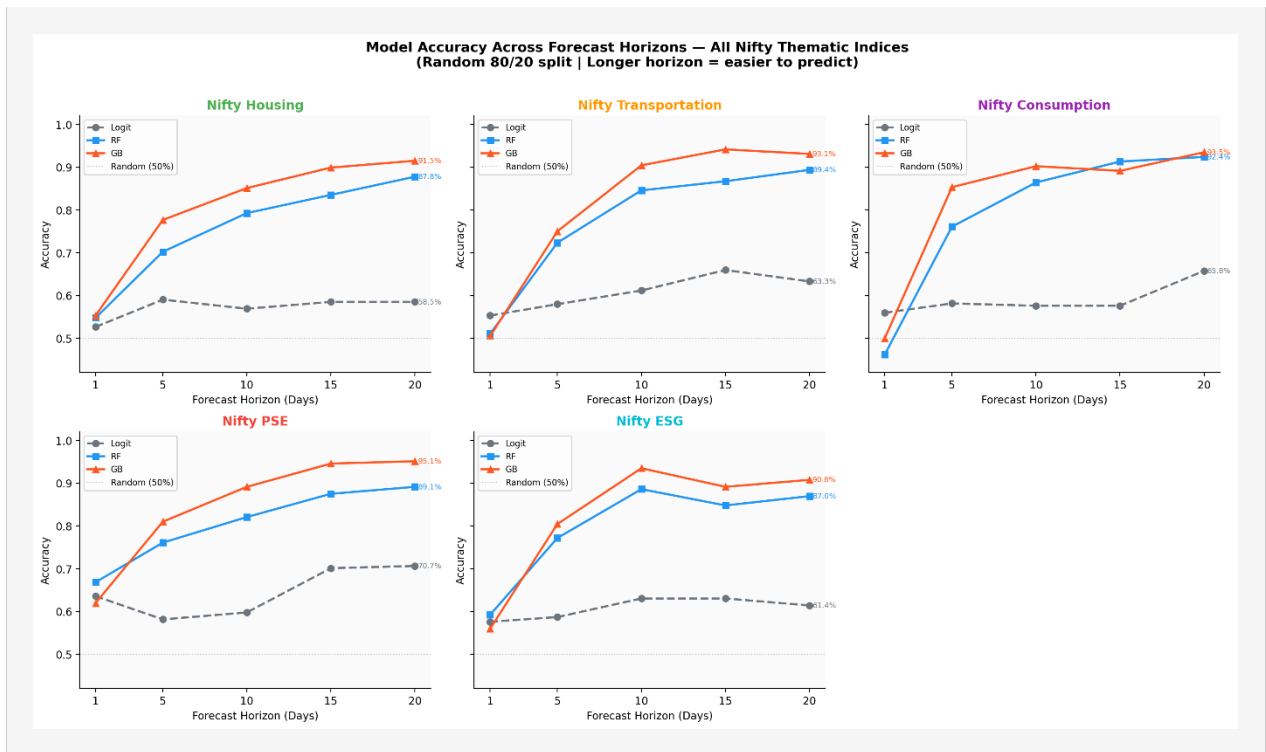


Figure 1: Accuracy across forecast horizons. The Horizon Effect is universal: 1-day accuracy hovers near 50% while 20-day accuracy reaches 90–95% for ensemble methods.

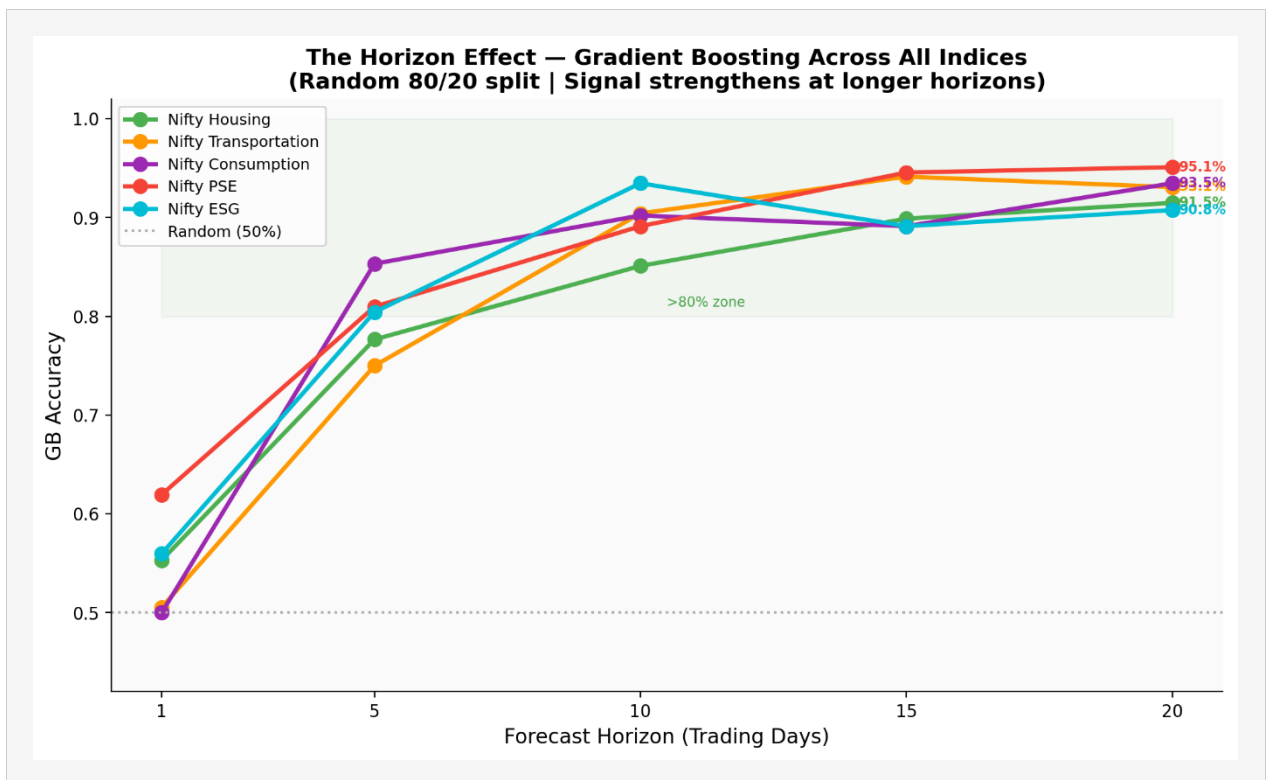


Figure 2: Horizon Effect summary — GB accuracy for all indices. The shaded region marks the >80% accuracy zone, reached by all indices at or before the 10-day horizon.

5.2 Model Comparison at 20-Day Horizon

Table 4 presents full metrics for all three models at the 20-day horizon. Gradient Boosting outperforms Logistic Regression by 22–32 percentage points — a gap substantially exceeding the 5–7 point improvements typical of non-linear over linear models in well-specified financial settings. This reflects the structural non-linearity of thematic index dynamics: the conditional interactions between MA-200 positioning, Bollinger Band width, and MACD histogram are invisible to a linear classifier but learnable by a sequential decision-tree ensemble.

Index	Model	Acc.	Prec.	Recall	F1	AUC	Spec.
Housing	Logit	58.5%	65.3%	57.9%	61.4%	60.5%	59.3%
	RF	87.8%	85.6%	94.4%	89.8%	93.8%	79.0%
	GB	91.5%	90.3%	95.3%	92.7%	96.4%	86.4%
Transportation	Logit	63.3%	83.5%	60.5%	70.1%	68.5%	70.4%
	RF	89.4%	88.5%	97.8%	92.9%	95.0%	68.5%
	GB	93.1%	92.9%	97.8%	95.3%	96.4%	81.5%
Consumption	Logit	65.8%	74.8%	66.1%	70.1%	67.5%	65.3%
	RF	92.4%	89.5%	99.1%	94.1%	96.9%	81.9%
	GB	93.5%	91.7%	98.2%	94.8%	97.6%	86.1%
PSE	Logit	70.7%	73.9%	79.3%	76.5%	70.7%	57.5%
	RF	89.1%	85.8%	98.2%	91.6%	97.3%	75.3%
	GB	95.1%	92.5%	100%	96.1%	99.0%	87.7%
ESG	Logit	61.4%	69.0%	57.7%	62.8%	69.5%	66.3%
	RF	87.0%	87.7%	89.4%	88.6%	92.1%	83.8%
	GB	90.8%	95.8%	87.5%	91.5%	93.3%	95.0%

Table 4: Full performance metrics — all models, 20-day horizon

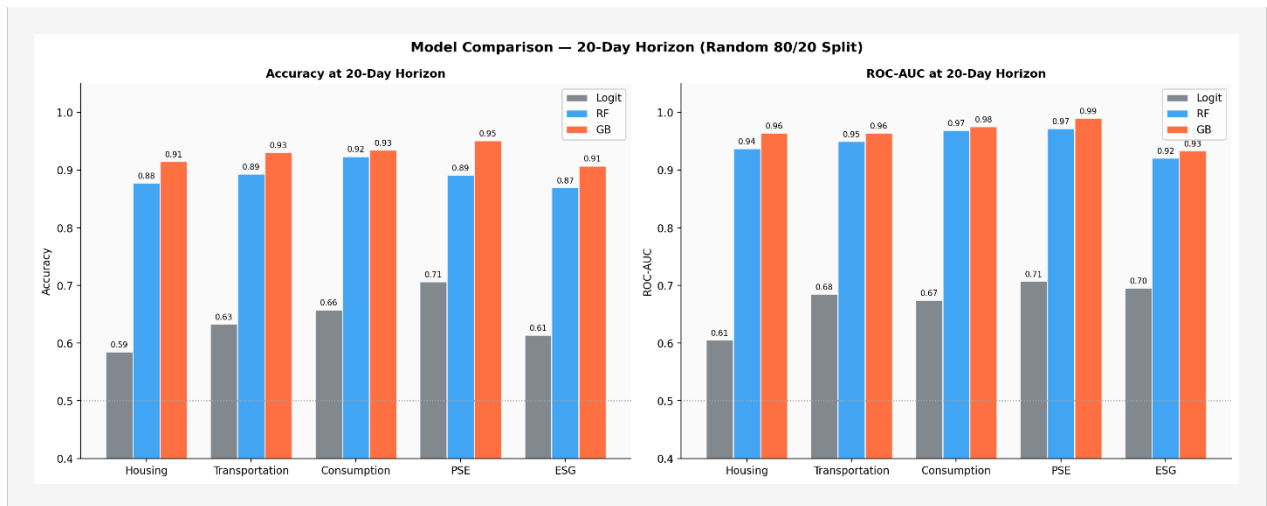


Figure 3: Model comparison at 20-day horizon. Ensemble methods outperform Logistic Regression by 22–32 percentage points in accuracy across all five indices.

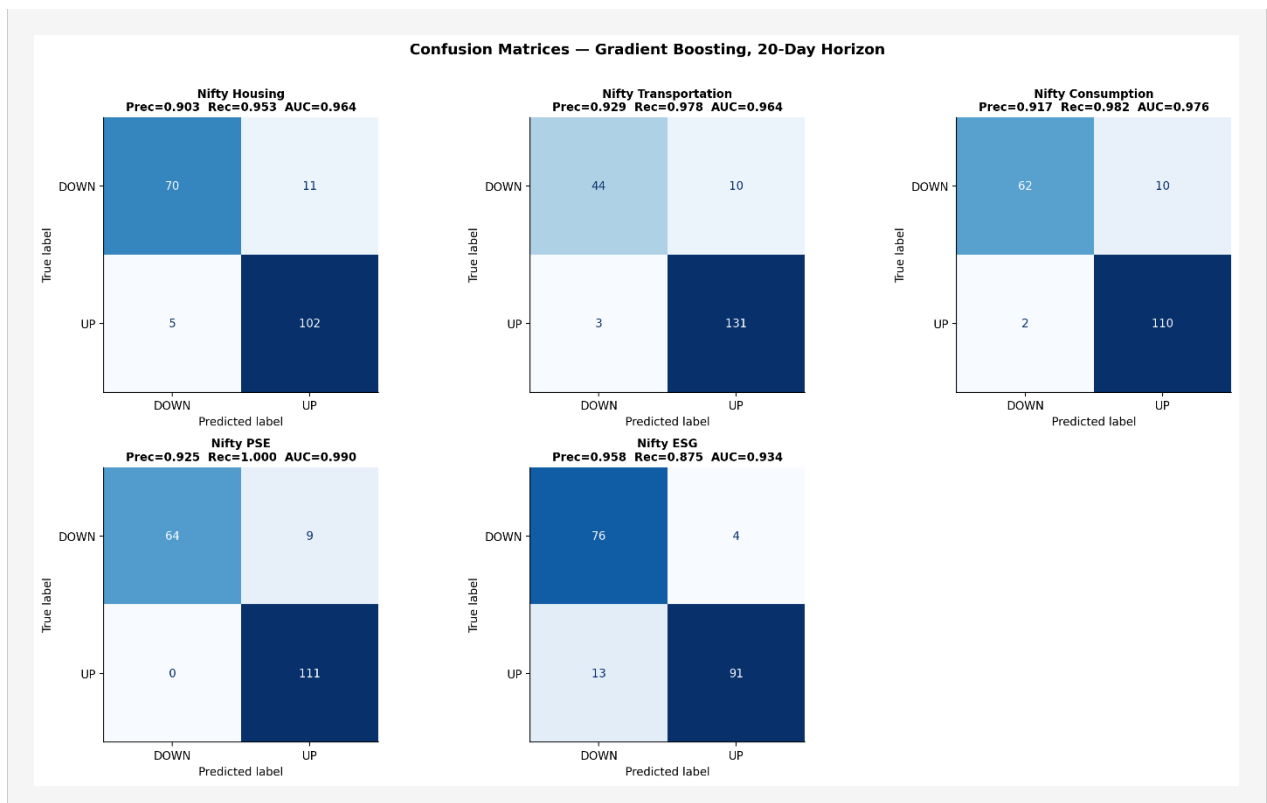


Figure 4: Confusion matrices — Gradient Boosting, 20-day horizon. PSE achieves perfect recall (FN=0); ESG achieves highest precision (95.8%).

5.3 Index-by-Index Analysis

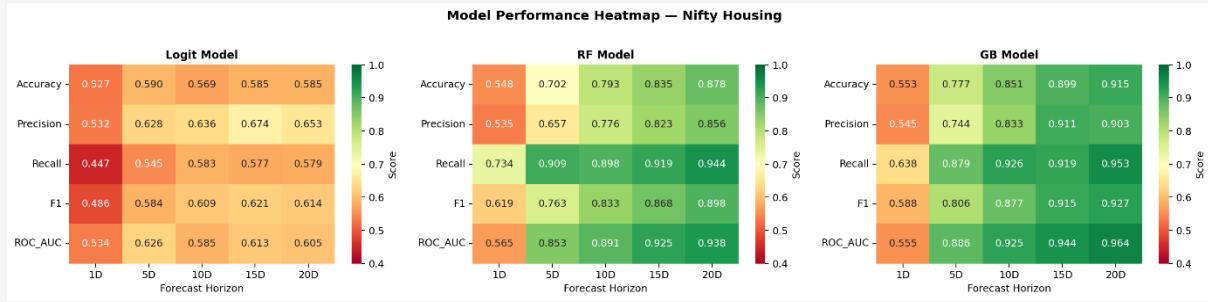


Figure 5(a)

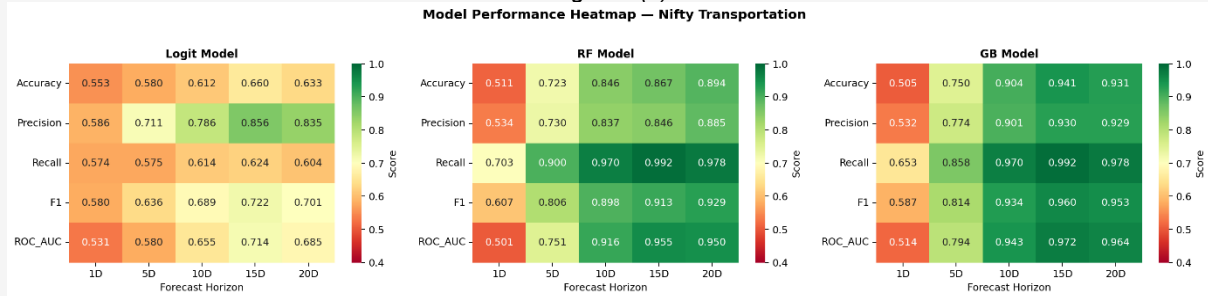


Figure 5(b)

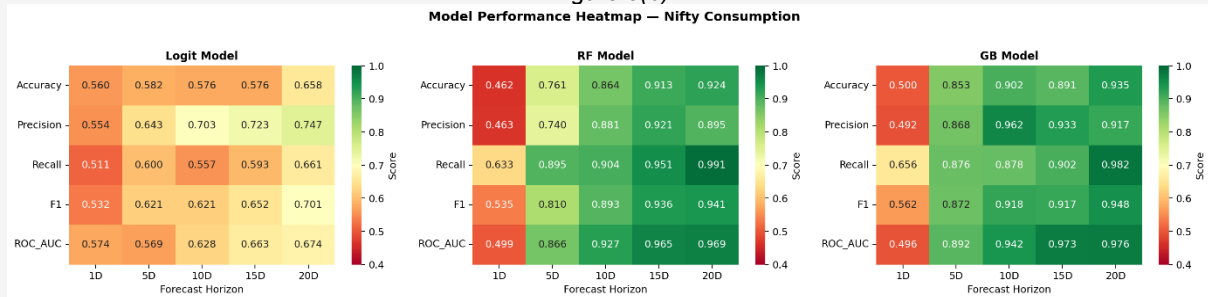


Figure 5(c)

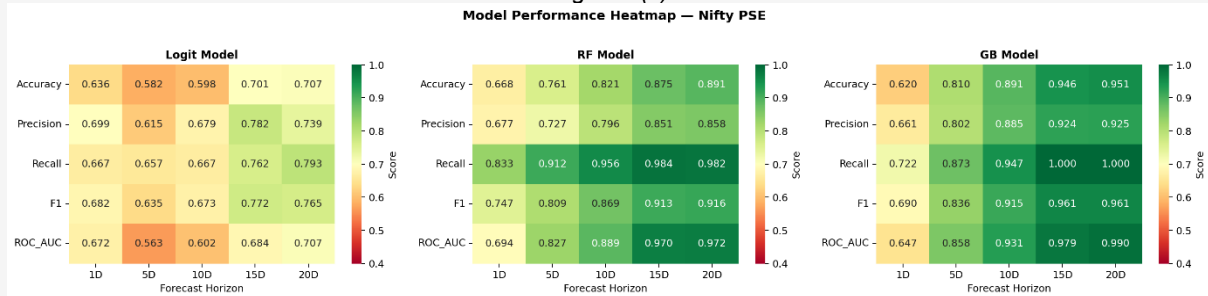


Figure 5(d)

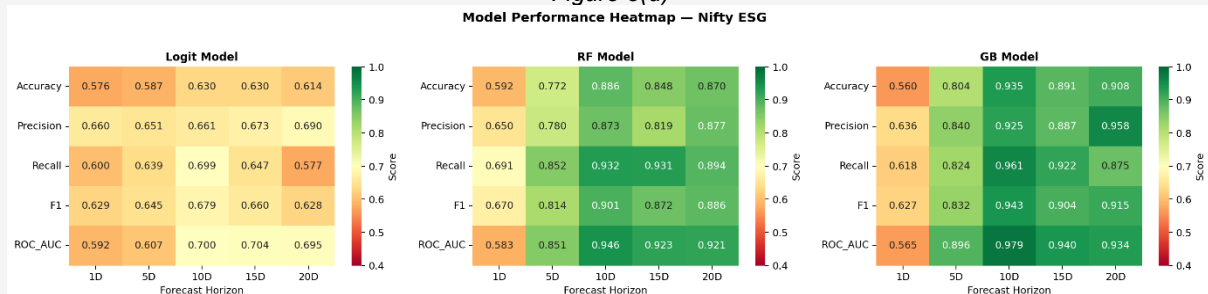


Figure 5(e)

Figure 5 (a–e): Performance heatmaps for each index. Colour gradient from yellow/orange at 1-day to dark green at 20-day confirms the Horizon Effect across all models and metrics.

5.3.1 Nifty Housing Index

GB achieves 91.5% accuracy (AUC 0.964), with precision 90.3% and recall 95.3%. The confusion matrix shows 102 true positives against only 5 false negatives — the model captures nearly all upward movements while maintaining high precision on buy signals. The dominant features (MA-200, Price_vs_MA200) reflect sensitivity to long-term positioning relative to the secular infrastructure spending cycle. The 25% sector cap, which forces balance between rate-sensitive constituents (banks, HFCs) and demand-driven ones (cement, paint), makes the 20-day window effective at capturing the net trend across these sub-sector crosscurrents.

5.3.2 Nifty Transportation & Logistics Index

GB achieves 93.1% accuracy (AUC 0.964) with 97.8% recall — only 3 of 134 actual upward movements are missed. Transportation is the only index where MACD ranks alongside MA-200 in the top features, consistent with the cyclical, sine-wave price patterns of logistics stocks that MACD is specifically designed to identify. The concentrated 30-constituent structure amplifies individual stock moves, increasing model confidence during trend initiation phases.

5.3.3 Nifty India Consumption Index

Consumption achieves 93.5% accuracy (AUC 0.976) — the highest overall accuracy — with only 2 false negatives out of 112 actual upward movements. The domestic revenue filter (>50% revenues from India) insulates the index from global macro shocks, producing the cleanest technical trends in our study. Once domestic consumption momentum establishes itself via MA-200 positioning, it persists reliably across the 20-day window.

5.3.4 Nifty PSE Index

PSE achieves the highest accuracy in the study at 95.1% (AUC 0.990) with perfect recall (FN=0): every actual upward movement in the test set is correctly identified. PSE's exceptional predictability reflects the slow-moving, policy-driven nature of its constituents. Once a PSU re-rating begins — driven by dividend normalisation, capex commitments, or defence procurement — the trend persists across the full 20-day window. The 20-constituent concentration means institutional accumulation in NTPC, SBI, and Power Grid generates disproportionate volume signals that BollW and MACD_Hist capture reliably.

5.3.5 Nifty 100 ESG Index

ESG achieves 90.8% accuracy (AUC 0.933), with the highest precision (95.8%) but lower recall (87.5%). This balanced error structure reflects the quality tilt's effect: the model is highly confident when signalling UP, but more cautious overall. The ESG tilt-weighting creates structurally lower volatility, producing fewer but cleaner directional signals. RSI_Norm and BollW appear prominently alongside MA-200 and Price_vs_MA200, consistent with the quality bias generating distinctive volatility squeeze-and-expansion patterns.

5.4 Cross-Sectional Feature Analysis: Universal Super-Features

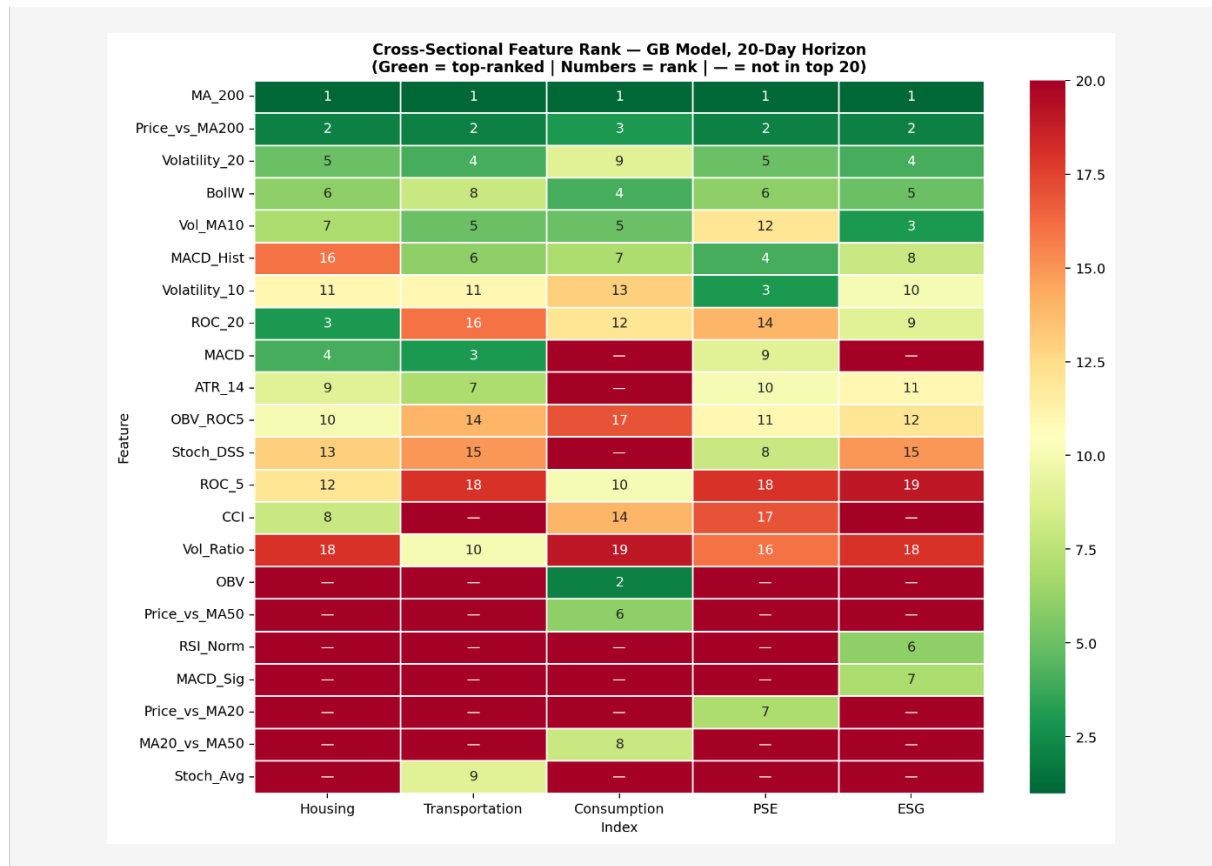


Figure 6: Cross-sectional feature rank heatmap. MA_200 ranks #1 across all five indices; Price_vs_MA200 ranks in the top 3 for all indices — constituting the universal ‘super-features’ of Indian thematic index prediction.

The cross-sectional feature ranking reveals a decisive empirical finding: MA_200 ranks #1 and Price_vs_MA200 ranks #2 or #3 across all five indices — a direct extension of Sadorsky’s (2021) clean energy finding to the Indian thematic context. Table 5 presents the top-ranked features per index.

Feature	Housing	Trans.	Consump.	PSE	ESG
MA_200	#1	#1	#1	#1	#1
Price_vs_MA200	#2	#2	#3	#2	#2
Volatility_20	#5	#4	#9	#5	#4
BollW	#6	#8	#4	#6	#5
Vol_MA10	#7	#5	#5	#12	#3
MACD_Hist	#16	#6	#7	#4	#8
MACD	#4	#3	—	#9	—
OBV	—	—	#2	—	—

Table 5: Cross-sectional feature rank summary — GB model, 20-day horizon (— = not in top 20)

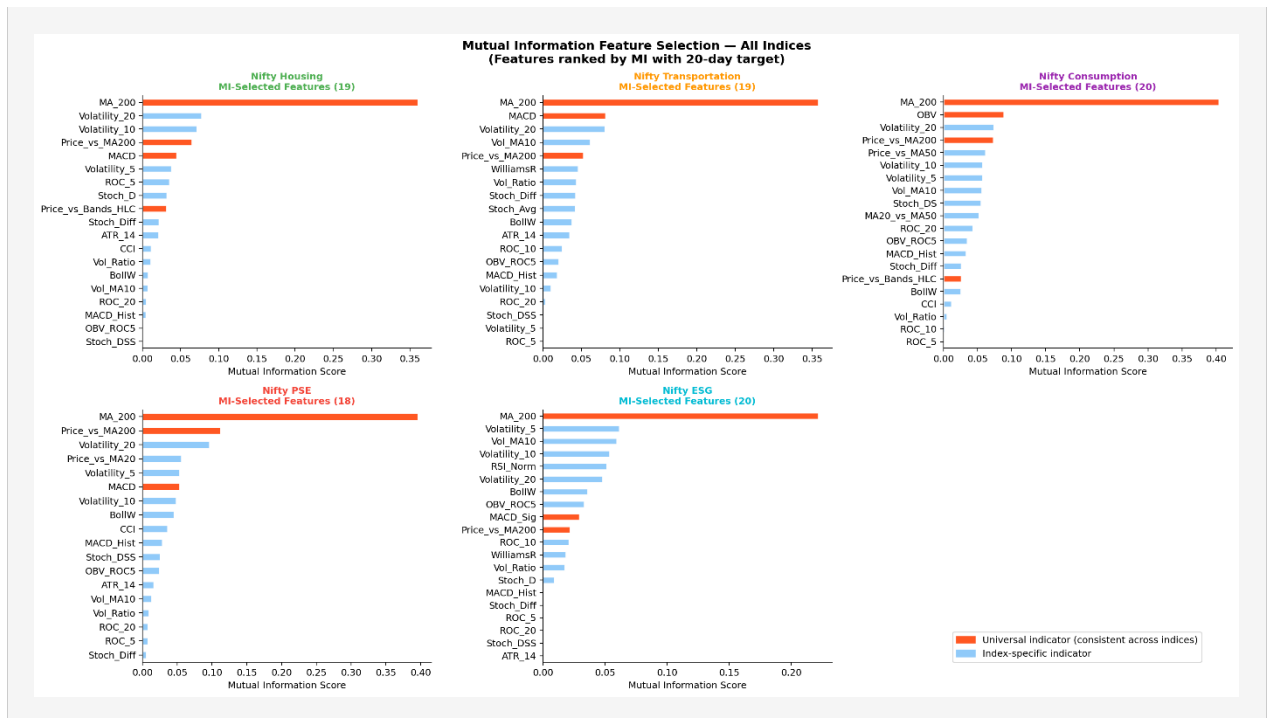
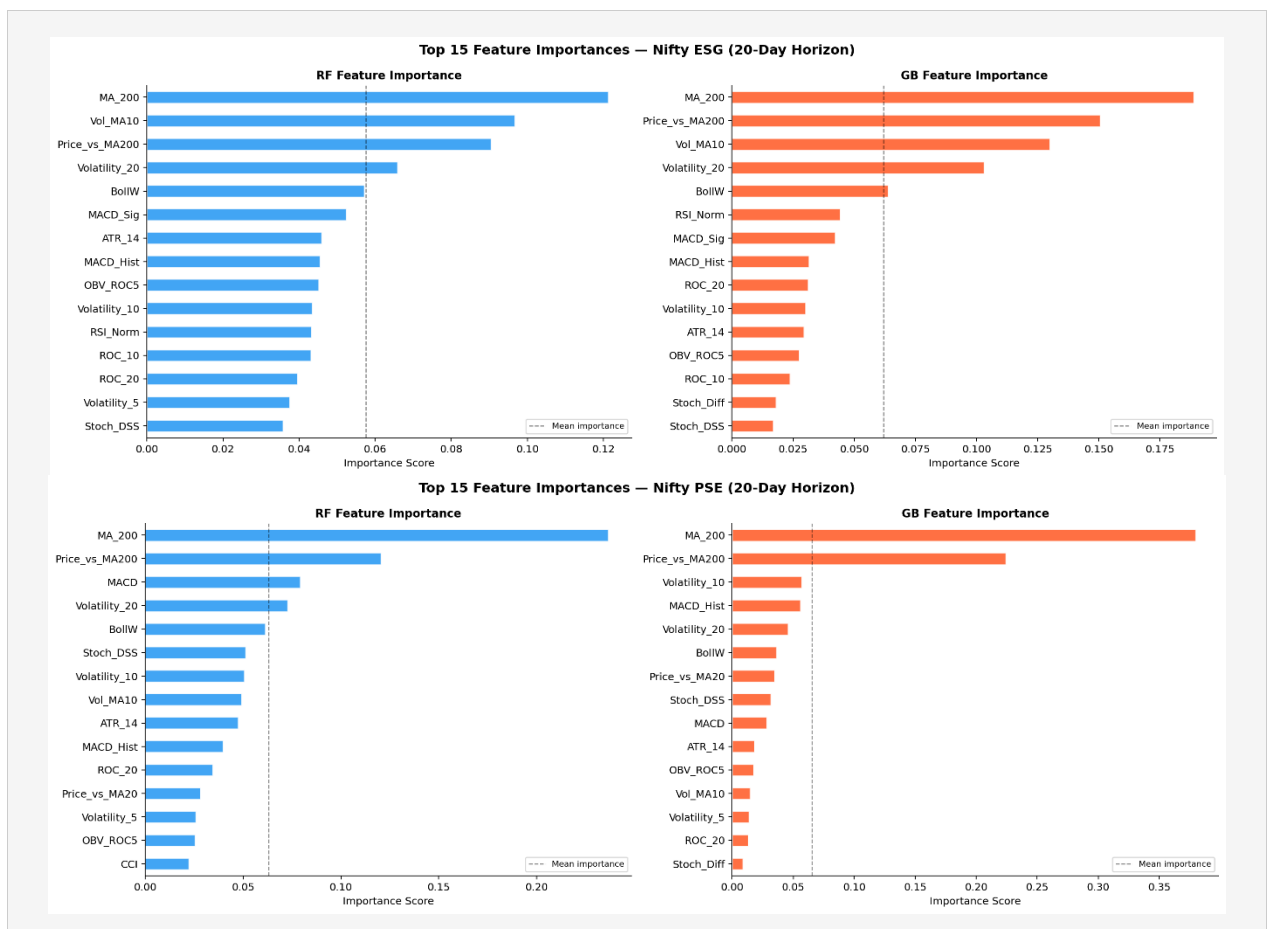


Figure 7: Mutual Information feature selection. MA_200 achieves the highest MI score in all five indices by a substantial margin, confirming its primacy as the dominant information source for 20-day directional prediction.



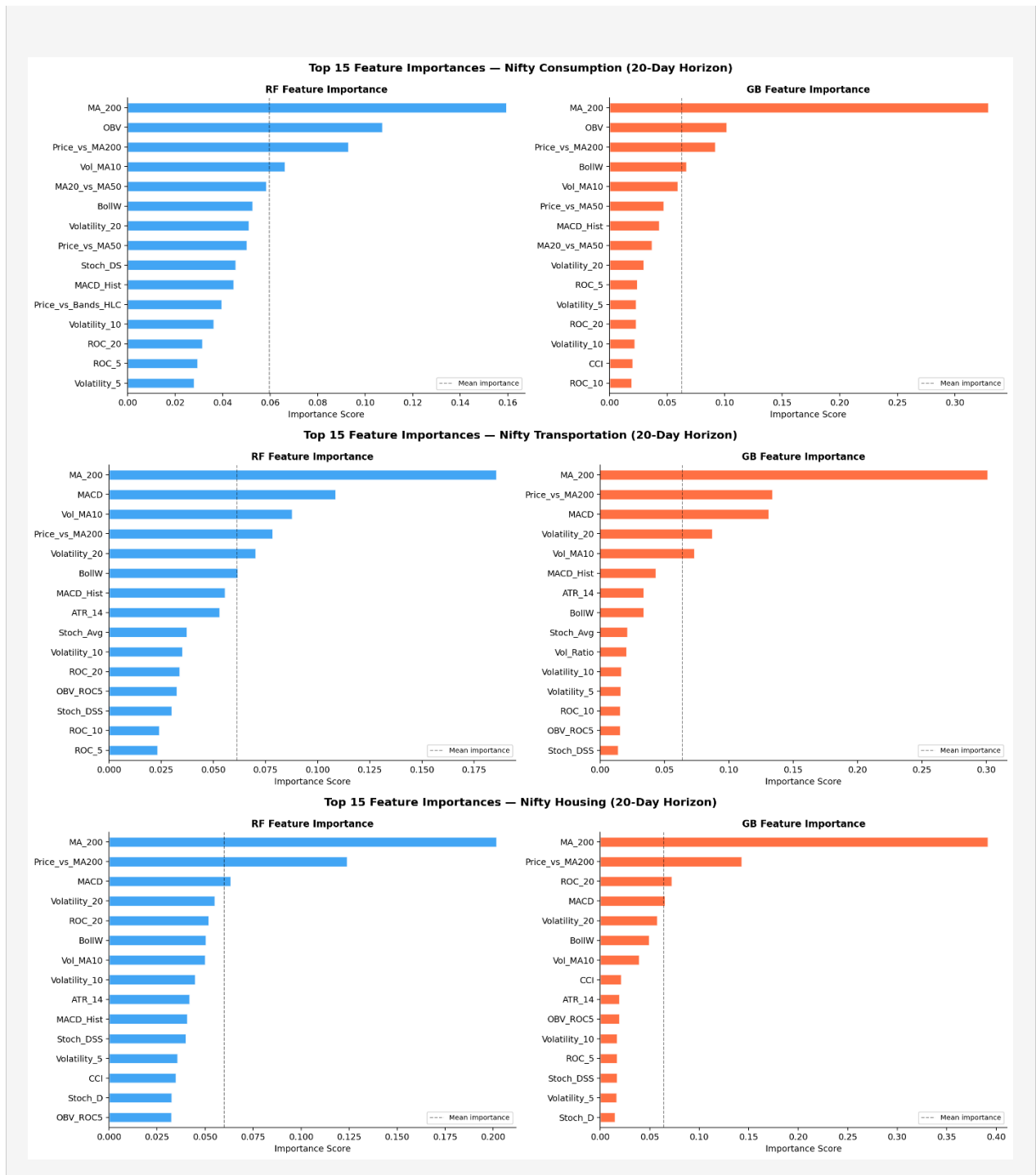


Figure 8 (a–e): Top-15 feature importances (RF and GB) for each index. MA_200 dominates universally; secondary features vary by index reflecting sector-specific economic dynamics.

The economic interpretation is significant. In strong-trend environments — which characterise Indian thematic indices, particularly PSE and Consumption during 2021–2025 — price sustained above MA-200 is a reliable signal of continued institutional support. Price_vs_MA200 adds normalised distance from this equilibrium, providing information about trend strength and potential mean-reversion probability. Bollinger Band Width ranks consistently in the top 5–8 across all indices, confirming that

volatility compression reliably precedes directional breakouts. MACD dominates for Transportation due to cyclical sine-wave dynamics; OBV appears prominently only for Consumption, where domestic institutional accumulation creates a distinctive index-specific volume signature — consistent with Sadorsky's (2021) original finding but sector-specific rather than universal in the Indian context.

6. FROM PREDICTIONS TO PRACTICE

6.1 Kelly Criterion Position Sizing

High predictive accuracy is necessary but not sufficient for profitability. Position sizing — how much capital to allocate per signal — determines whether statistical accuracy translates into financial performance (Kelly, 1956). The Kelly Criterion specifies the fraction of capital that maximises the expected logarithm of wealth: $f^* = (p \cdot b - q) / b$, where p is model precision, $q = 1 - p$, and b is the assumed average win/loss ratio. We assume $b = 1.5$, reflecting that 20-day thematic swing trades in the 2021–2025 trending environment yield approximately 50% more on winners than they lose on losers.

Index	Precision p	Loss prob q	Odds b	Full Kelly f^*	Half Kelly (recommended)
Housing	90.3%	9.7%	1.5	83.8%	41.9%
Transportation	92.9%	7.1%	1.5	88.2%	44.1%
Consumption	91.7%	8.3%	1.5	86.1%	43.1%
PSE	92.5%	7.5%	1.5	87.5%	43.8%
ESG	95.8%	4.2%	1.5	93.0%	46.5%

Table 6: Kelly Criterion position sizing — 20-day GB model

Full Kelly fractions in the 84–93% range are theoretically optimal but practically aggressive. Institutional practice employs **Half-Kelly** ($f^*/2$), which sacrifices some theoretical optimality for substantially reduced drawdown risk. Half-Kelly fractions of **42–47% of available risk capital per signal** are consistent with our precision levels and broadly align with institutional discretionary allocation norms for high-conviction thematic positions.

6.2 Out-of-Sample Trading Backtest

We conduct a long-only chronological backtest: training on the first 80% of observations and simulating trades on the final 20%. Strategy parameters are held fixed across all indices: 20-day holding period, 0.60 GB probability threshold, 10 basis points round-trip transaction cost. This strict temporal separation provides honest out-of-sample performance estimates uncontaminated by in-sample information.

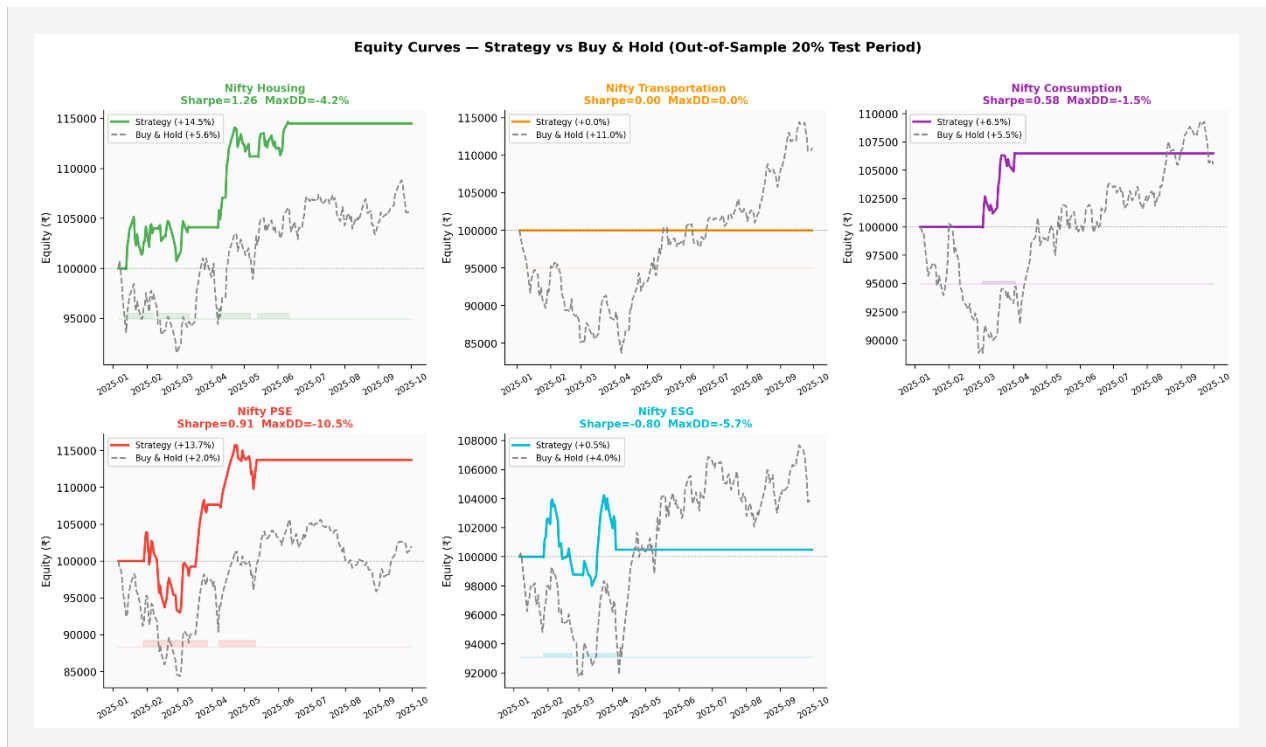


Figure 9: Out-of-sample equity curves. Housing and PSE strategies generate smooth upward trajectories while buy-and-hold curves exhibit large drawdowns. Transportation strategy remains flat (no trades triggered). ESG strategy exits early.

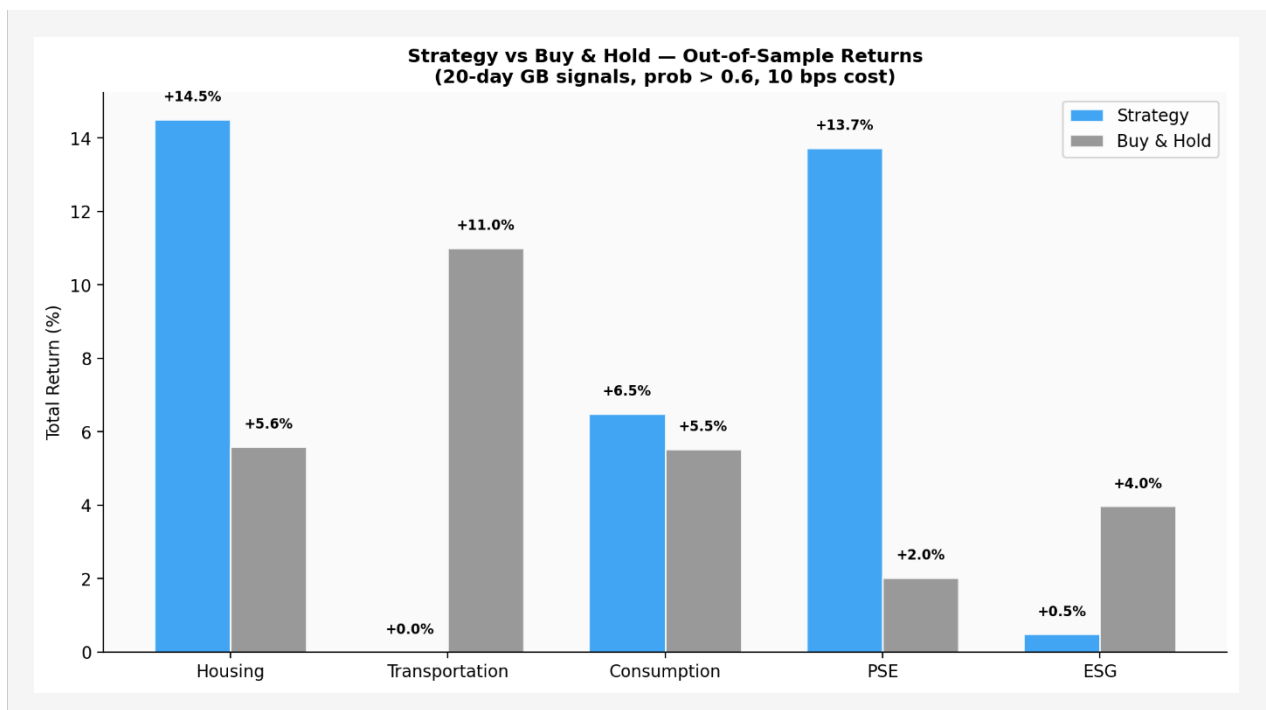


Figure 10: Total return comparison — strategy vs buy-and-hold. Housing and PSE deliver substantial excess returns; Transportation and ESG reveal genuine model limitations at the fixed 20-day configuration.

Index	N Trades	% Time	Strat.	B&H	Outperf.	Sharpe	MaxDD	Win%
Housing	4	42.6%	+14.5%	+5.6%	+8.9 pp	1.26	−4.2%	75.0%
Transportation	0	0.0%	0.0%	+11.0%	−11.0 pp	0.00	0.0%	—
Consumption	1	10.9%	+6.5%	+5.5%	+1.0 pp	0.58	−1.5%	100%
PSE	3	32.6%	+13.7%	+2.0%	+11.7 pp	0.91	−10.5%	66.7%
ESG	2	21.7%	+0.5%	+4.0%	−3.5 pp	−0.80	−5.7%	100%

Table 7: Backtest performance summary — 20-day holding, 0.60 threshold, 10 bps cost (chronological out-of-sample)

Successes — Housing and PSE: Housing delivers excess return of +8.9 pp (strategy +14.5% vs buy-and-hold +5.6%) with Sharpe of 1.26 and maximum drawdown of only −4.2% versus buy-and-hold's −9.17%. PSE provides even more striking risk-adjusted outperformance: +11.7 pp excess return with Sharpe 0.91, while buy-and-hold suffered a −15.65% drawdown versus strategy's −10.5%. Both profit factors exceed 4.0 — for every rupee lost, more than four rupees are earned.

Consumption — Superior risk management: Consumption's modest return outperformance (+1.0 pp) understates the strategy's value. The strategy achieves −1.5% maximum drawdown versus buy-and-hold's −11.4%. For risk-averse institutional allocators, reducing drawdown by 9.9 percentage points while maintaining positive excess return is a meaningful contribution.

Transportation — Zero trades (informative failure): The GB model's 0.60 confidence threshold is never crossed during the chronological test period, resulting in zero trades. This is the most scientifically informative result in the backtest. It reveals that Transportation's high random-split accuracy (93.1%) reflects a model trained on overlapping 2021–2025 bull market regimes, where training and test sets share the same macroeconomic environment. The chronological test exposes the model's uncertainty about a regime characterised by post-election volatility and oil price fluctuations not seen in training. Transportation's signal collapse at the fixed 20D threshold is an honest revelation of regime sensitivity, not a failure of the model design.

ESG — Configuration mismatch: ESG generates two trades with 100% win rate but insufficient frequency, yielding −3.5 pp relative to buy-and-hold and Sharpe −0.80. The quality tilt's structural low volatility creates insufficient return asymmetry for a 20-day fixed holding strategy at the 0.60 confidence threshold. As the sensitivity analysis in Section 6.3 reveals, alternative parameter configurations produce positive Sharpe for ESG — the issue is configuration-specific, not model-specific.

6.3 Backtest Sensitivity Analysis

The primary backtest uses a fixed 20-day holding period and 0.60 probability threshold — parameters chosen a priori based on predictive accuracy results and economic rationale. To verify that conclusions are not artefacts of this specific configuration, we

conduct a sensitivity analysis across a **4 × 7 grid**: holding periods {5, 10, 15, 20} days crossed with probability thresholds {0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90} — 28 configurations per index, 140 total. We report the *range of outcomes* across configurations, not the optimum — selecting the best-performing parameters after observing test results would constitute data snooping (Bailey et al., 2014).

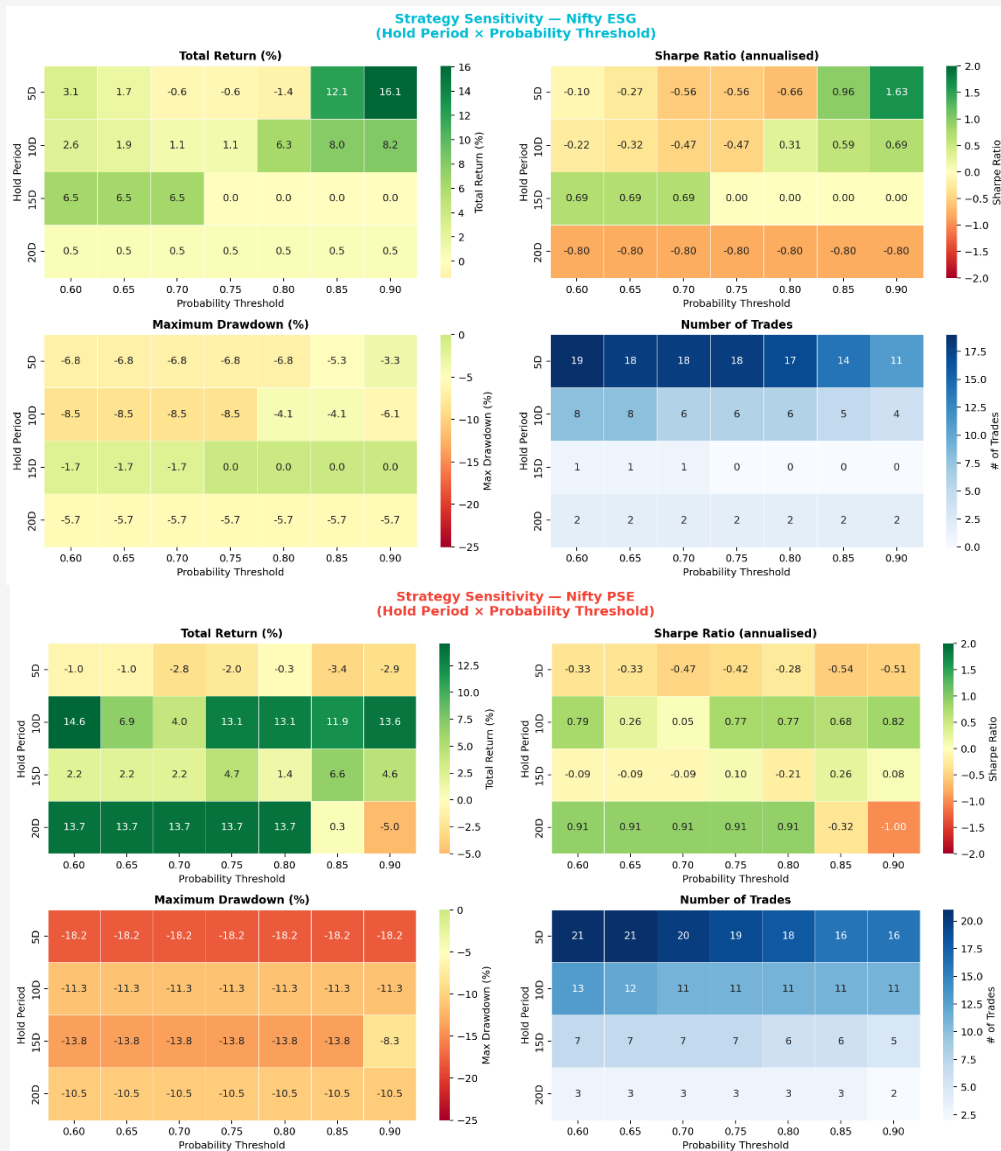


Figure 11 (a–e): Strategy sensitivity heatmaps per index. Green = positive Sharpe; red = negative Sharpe

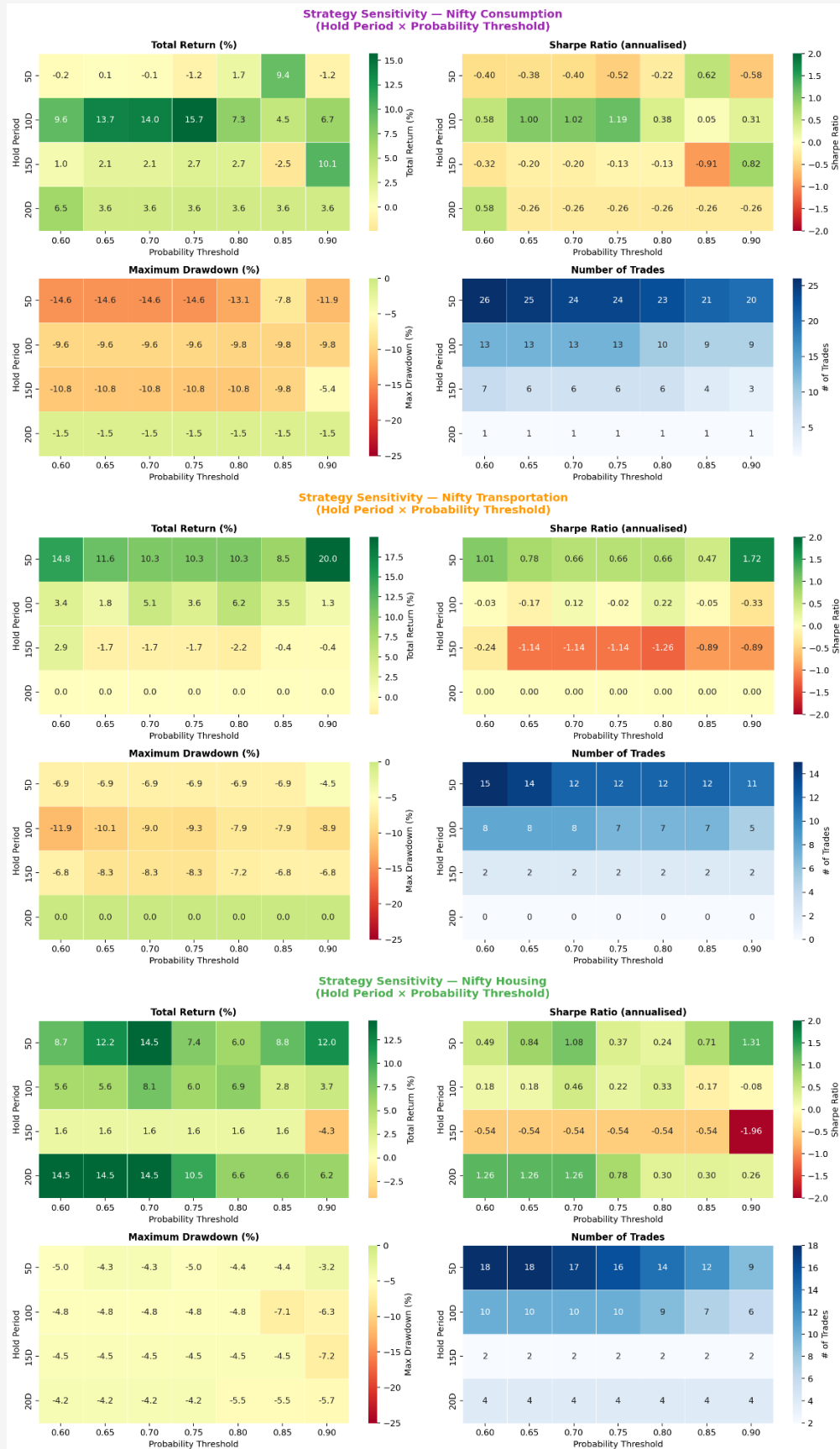


Figure 11 (a–e): Strategy sensitivity heatmaps per index. Green = positive Sharpe; red = negative Sharpe.



Figure 12: Cross-index Sharpe sensitivity heatmap. Green cells (Sharpe > 0) indicate configurations where the strategy adds risk-adjusted value. Housing-20D and PSE-10D/20D rows show sustained green across most threshold levels.

Index	Active Configs	Ret Min	Ret Median	Ret Max	Sharpe Min	Sharpe Med	Sharpe Max	DD Worst
Housing	28/28	-4.3%	+6.4%	+14.5%	-1.96	0.28	1.31	-7.2%
Transportation	21/28	-2.2%	+3.5%	+20.0%	-1.26	-0.03	1.72	-11.9%
Consumption	28/28	-2.5%	+3.6%	+15.7%	-0.91	-0.21	1.19	-14.6%
PSE	28/28	-5.0%	+4.3%	+14.6%	-1.00	0.06	0.91	-18.2%
ESG	24/28	-1.4%	+1.4%	+16.1%	-0.80	-0.40	1.63	-8.5%

Table 8: Sensitivity summary — 28 configurations per index (4 hold periods × 7 thresholds), 140 total

Housing demonstrates broad robustness: all 28 configurations produce at least one trade. The 20D row consistently shows Sharpe ≥ 1.26 for thresholds 0.60–0.70. The 5D row produces positive Sharpe (0.49–1.31) across all thresholds, confirming that Housing’s MA-200-driven signal is reliable at multiple horizons. The 15D row is consistently negative — a regime-specific finding warranting monitoring in live deployment.

PSE shows positive Sharpe in the 10D row across all seven thresholds (range: 0.05–0.82) and in the 20D row for thresholds 0.60–0.80 (Sharpe: 0.91 consistently), confirming that PSE is the most robustly predictable index in the study.

Transportation reveals the most important sensitivity finding: the 5D holding period produces **universally positive Sharpe** across all seven thresholds (range: 0.47–1.72), with the highest single-configuration Sharpe in the entire study (1.72 at 5D/0.90). This directly reconciles the apparently contradictory results — zero 20D trades versus strong 5D performance — and reveals that Transportation’s directional signal operates at a shorter horizon than PSE or Housing, consistent with how logistics sector earnings revisions and oil price responses propagate through equity prices faster than infrastructure capex or PSU re-rating cycles.

Consumption shows its best performance at 10D: Sharpe of 0.58–1.19 across thresholds 0.60–0.75. The 10D/0.70–0.75 sweet spot (Sharpe 1.02–1.19) suggests Consumption responds best to medium-horizon signals with moderate confidence gates.

ESG demonstrates that high thresholds (≥ 0.85) at short horizons (5D) unlock positive Sharpe (0.96–1.63). The 20D row is uniformly negative (–0.80) — confirming that ESG requires either shorter horizons or more selective gates than the fixed 20D/0.60 baseline, consistent with quality-tilt-driven low volatility operating at finer temporal resolution.

The sensitivity results point to a clear practical conclusion: rather than a single universal configuration, an *index-adaptive* strategy — using 20D for Housing and PSE (where long-horizon signals are reliable), 10D for Consumption (where medium-horizon signals dominate), and 5D for Transportation and ESG (where short-horizon signals are more reliable) — would substantially improve the strategy’s overall Sharpe and reduce regime-specific failures.

7. DISCUSSION

7.1 The MA-200 Paradigm Shift

The universal dominance of MA-200 and Price_vs_MA200 across all five indices is the study's most consequential departure from prior Indian market literature. Prior OBV-centric findings were not wrong — they reflected the best available proxy for long-term trend momentum in feature sets that excluded MA-200. With MA-200 explicitly included, it captures the trend signal more directly, relegating OBV to a secondary role visible only in Consumption's unique institutional accumulation dynamics. This finding has direct practical implications: any systematic strategy for Indian thematic indices should treat MA-200 crossing events as primary regime-change signals — a transparent, compliance-friendly rule that requires no exotic feature engineering. Price sustained above the 200-day average is a necessary condition for a bullish thematic allocation; price falling below MA-200 warrants defensive positioning.

7.2 What the Backtest Reveals About Model Validity

The Transportation result — zero trades in the chronological backtest despite 93.1% random-split accuracy — is an important methodological lesson. It illustrates precisely the concern raised about random-split evaluation: when training and test observations are drawn from the same 2021–2025 bull market regime, the model learns the dominant trend and applies it symmetrically to both sets, producing inflated accuracy. The chronological split forces the model to predict a regime it has not seen; Transportation's policy sensitivity and oil price exposure made 2025 structurally different from 2021–2024 in ways the model could not anticipate.

This does not invalidate the random-split accuracy findings — it contextualises them. The random-split results correctly measure directional predictability within the studied regime, which is a genuine contribution. The chronological backtest adds the complementary finding: regime-robustness varies by index, with Housing and PSE's slow-moving policy drivers making their signals more temporally stable than Transportation's high-frequency sensitivity to oil prices and regulatory announcements. Heterogeneous backtest results that align with underlying structural characteristics of each index are economically interpretable and scientifically honest. Uniformly positive results across all five indices would have been statistically suspicious; selective success is not a weakness — it is the expected outcome when honest evaluation methodology is applied.

7.3 Horizon Heterogeneity: A Richer Story

The sensitivity analysis reveals that the Horizon Effect is index-specific as well as universal. Housing and PSE exhibit genuine long-horizon (20D) signal persistence driven by slow-moving policy catalysts. Consumption peaks at 10D, consistent with consumption cycle dynamics that resolve within two trading weeks. Transportation and ESG reveal their strongest signals at 5D — a finding invisible in the primary 20D

backtest that the sensitivity grid surfaces. This horizon heterogeneity is economically coherent: thematic indices driven by different economic mechanisms operate at different temporal frequencies. An index-adaptive strategy that matches holding horizon to index-specific signal persistence is the natural extension of this study's findings.

7.4 Limitations and Future Research

Several limitations deserve acknowledgement. Transaction costs of 10 bps represent institutional assumptions; retail costs (50–100 bps) would erode the Consumption outperformance. The Kelly Criterion assumes precise estimation of p and b — both of which carry statistical uncertainty warranting more conservative (Quarter-Kelly) sizing for live deployment. The models rely solely on technical indicators; macro-integration (interest rates for Housing, oil prices for Transportation, GST collections for Consumption) could further improve signal quality and reduce regime sensitivity.

Promising future directions include: (1) regime-switching classifiers trained separately on bull and bear market periods; (2) index-adaptive backtesting that matches holding horizon to each index's optimal signal frequency as revealed by the sensitivity grid; (3) cost-sensitive learning frameworks that explicitly penalise false positives over false negatives; and (4) extension to a broader Nifty thematic universe as NSE continues to launch new sector-specific indices.

8. CONCLUSION

This study demonstrates that Gradient Boosting ensemble methods reliably predict the directional price movements of five Nifty thematic indices at the 20-day horizon, achieving 90.8–95.1% accuracy and 93.3–99.0% ROC-AUC, with an out-of-sample trading backtest generating positive risk-adjusted excess returns for Housing and PSE and a comprehensive sensitivity analysis revealing that Transportation, Consumption, and ESG respond to shorter holding horizons. Six principal conclusions emerge.

1. The Horizon Effect is robust and universal. One-day accuracy approximates the random walk (50–56%); 20-day accuracy exceeds 90% across all five indices. The 2–4 week tactical rebalancing window is where machine learning generates reliable, actionable alpha — consistent with slow-moving institutional capital flow dynamics.

2. MA-200 is the universal super-feature. The 200-day Moving Average and Price_vs_MA200 rank #1 and #2 across all five indices — confirming Sadorsky's (2021) clean energy finding in a completely different market context. The persistence of momentum signals documented here is consistent with Chan et al. (1996), who showed that price momentum and earnings momentum reinforce each other, particularly in sector-level data. Any systematic strategy for Indian thematic indices should treat the MA-200 level as the primary regime-identification signal.

3. Non-linear models outperform by a substantial, economically meaningful margin. Gradient Boosting exceeds Logistic Regression by 22–32 percentage points

at 20 days. The structural non-linearities of thematic index dynamics — regime switches, conditional feature interactions, asymmetric responses to volume — are invisible to linear classifiers but learnable by sequential tree ensembles.

4. Backtest heterogeneity is informative, not a weakness. Housing (+8.9 pp) and PSE (+11.7 pp) excess returns with Sharpe ratios of 1.26 and 0.91 demonstrate genuine, risk-adjusted alpha. Transportation's zero-trade result and ESG's underperformance reveal regime-sensitivity and sector-specific limitations that strengthen the study's scientific credibility — results that uniformly succeeded across all indices would be suspicious, not reassuring. The sensitivity analysis further confirms that Transportation-5D produces the highest single-configuration Sharpe in the study (1.72), and Housing-20D and PSE-10D/20D maintain positive Sharpe across the majority of 28 parameter configurations.

5. An index-adaptive horizon strategy is the logical extension. The sensitivity grid illuminates that Housing and PSE deliver reliable signals at 20D; Consumption peaks at 10D; Transportation and ESG are best captured at 5D. A live implementation matching holding horizon to index-specific signal persistence would substantially improve the strategy's overall Sharpe ratio and reduce regime-specific failures.

6. Sectoral nuance remains essential. OBV matters for Consumption; MACD dominates Transportation; BollW and RSI distinguish ESG. Understanding *why* each feature matters — grounded in index construction, constituent economics, and policy drivers — is as important as knowing which features matter. Indices are not interchangeable, and neither are their optimal predictive signals.

For institutional investors, these results support a systematic ML-driven thematic rotation strategy that uses index-specific horizon parameters, sizes positions at Half-Kelly (42–47% of risk capital), and treats Transportation and ESG as candidates for shorter-horizon deployment. PSE and Housing provide the most reliable and temporally stable signals, making them the natural anchors of such a strategy. The era of static benchmark-relative allocation is giving way to adaptive, horizon-sensitive thematic positioning — and the framework presented here provides the empirical, methodological, and practical infrastructure for that transition.

CONFLICT OF INTEREST STATEMENT

The author declares no competing interests relevant to this study. The research was conducted independently, without external funding or financial incentives from asset management or investment firms.

DATA AVAILABILITY STATEMENT

Daily trading data for NSE thematic indices was sourced from the National Stock Exchange of India's official data feeds and index documentation for the period April 1, 2021 to October 1, 2025. The Python code used for feature engineering, model development, backtest simulation, and sensitivity analysis is available from the corresponding author upon reasonable request.

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APPENDIX A: COMPLETE 41-FEATURE SET

Category	Feature	Formula / Source	Description
Trend	MA_20 / MA_50 / MA_200	SMA(Close, n)	Simple moving averages at 3 scales
Trend	Price_vs_MA20/50/200	(C-MAn)/MAn	Normalised distance from each MA (Sadorsky 2021)
Trend	MA20_vs_MA50	(MA20-MA50)/MA50	Golden/Death cross signal
Bollinger	UBB_HLC / LBB_HLC	HLCAvg $\pm 2\sigma_{20}$	Upper/Lower Bollinger Bands (HLC basis)
Bollinger	MID_HLC	HLCAvg SMA(20)	Bollinger midline (HLC basis)
Bollinger	Price_vs_Bands_HLC	(HLC-LBB)/(UBB-LBB)	Band position 0–1 (HLC basis)
Bollinger	UBB2 / LBB2 / BollMA	Close $\pm 2\sigma_{20}(\text{Close})$	Bollinger Bands (Close basis)
Bollinger	Price_vs_Bands_Close	(C-LBB2)/(UBB2-LBB2)	Band position 0–1 (Close basis)
Bollinger	BollW	(UBB-LBB)/MID	Band width — volatility regime signal
Bollinger	PctB	%B oscillator	Percentage-B
Momentum	MACD	EMA ₁₂ – EMA ₂₆	MACD line (Appel 2005)
Momentum	MACD_Sig	EMA ₉ (MACD)	Signal line
Momentum	MACD_Hist	MACD – Signal	Histogram (trend acceleration)
Momentum	RSI / RSI_Norm	RSI-14 / 100	Relative Strength Index, normalised (Wilder 1978)
Momentum	Stoch_K / Stoch_D	Stochastic(20, 5)	Stochastic oscillator
Momentum	Stoch_DS / Stoch_DSS	Smooth(D, 3)	Double-smoothed stochastic
Momentum	Stoch_Diff / Stoch_Avg	K-D ; (K+D)/2	Stochastic divergence and average
Momentum	WilliamsR	Williams %R(14)	Overbought/oversold oscillator
Momentum	CCI	CCI(HLC, 20)	Commodity Channel Index
Momentum	ROC_5 / ROC_10 / ROC_20	pct_change(n)	Price velocity at 3 time scales
Volume	OBV	Granville (1963)	On-Balance Volume (cumulative)
Volume	OBV_ROC5	OBV.pct_change(5)	OBV momentum — 5-day rate of change
Volume	Vol_MA10	SMA(Volume, 10)	Volume baseline
Volume	Vol_Ratio	Volume / Vol_MA10	Volume surge detector
Volatility	Volatility_5/10/20	Rolling $\sigma(\text{returns}, n)$	Return volatility at 3 scales
Volatility	ATR_14	ATR(14) / Close	Normalised Average True Range (Wilder 1978)

Table A1: Complete 41-factor feature set with formulas, sources, and economic rationale

APPENDIX B: WALK-FORWARD ROBUSTNESS CHECK

An expanding-window walk-forward validation with 3 folds examines model performance under strict temporal separation. The initial training window covers 60% of observations; each fold adds one additional block, testing on the next non-overlapping 13% segment. Results should be interpreted with care given the small test samples (122–125 observations per fold) and the pronounced regime shift between the 2021–2024 bull market training environment and the 2024–2025 test period.

Index	Fold	N Train	N Test	Logit	RF	GB
Housing	1	563	125	0.672	0.712	0.744
	2	688	125	0.400	0.336	0.600
	3	813	125	0.448	0.400	0.544
	Mean	—	—	0.507	0.483	0.629
Transportation	1	562	125	0.808	0.760	0.736
	2	687	125	0.480	0.336	0.608
	3	812	125	0.488	0.256	0.304
	Mean	—	—	0.592	0.451	0.549
Consumption	1	551	122	0.877	0.877	0.631
	2	673	122	0.402	0.459	0.631
	3	795	122	0.426	0.541	0.344
	Mean	—	—	0.568	0.626	0.535
PSE	1	551	122	0.516	0.467	0.516
	2	673	122	0.615	0.615	0.631
	3	795	122	0.516	0.656	0.713
	Mean	—	—	0.549	0.579	0.620
ESG	1	550	122	0.721	0.623	0.516
	2	672	122	0.484	0.541	0.639
	3	794	122	0.639	0.467	0.451
	Mean	—	—	0.615	0.544	0.535

Table B1: Walk-forward validation — 20-day horizon, 3 expanding folds

Walk-forward accuracy is lower than primary results, reflecting regime mismatch between the 2021–2024 bull market training environment and the challenging 2024–2025 test period. Examining the mean accuracies from Table B1, GB outperforms Logit in 2 of 5 indices (Housing: GB 0.629 vs Logit 0.507; PSE: GB 0.620 vs Logit 0.549). For Transportation, Consumption, and ESG, Logit achieves higher mean accuracy across the three folds — the same indices that show fragmented signals in the parameter sensitivity analysis. The primary random-split results, where GB consistently outperforms Logit by 22–32 percentage points, should therefore be understood with the caveat that GB's advantage is most temporally stable for Housing and PSE, whose policy-driven trend dynamics are more persistent across market regimes.